

Homework 7

Due: Sunday, 21st July 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

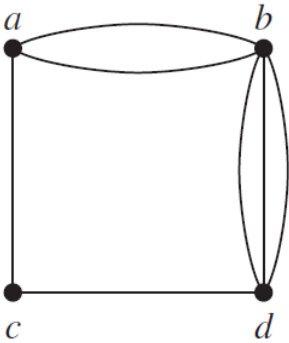
Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

For Questions 1 and 2, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph in the table shown in class (Graphs_intro, page 25).

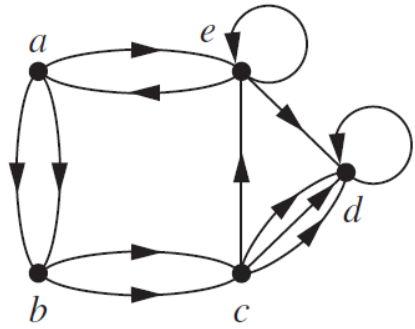
Question 1.

Describe the type of the following graph.



Question 2.

Describe the type of the following graph.



Question 3.

For each graph in Questions 1 and 2 that is not simple, find a set of edges to remove to make it simple. Keep the size of the set as small as possible.

Question 4.

Draw the acquaintanceship graph that represents that Tom and Patricia, Tom and Hope, Tom and Sandy, Tom and Amy, Tom and Marika, Jeff and Patricia, Jeff and Mary, Patricia and Hope, Amy and Hope, and Amy and Marika know each other, but none of the other pairs of people listed know each other.

Question 5.

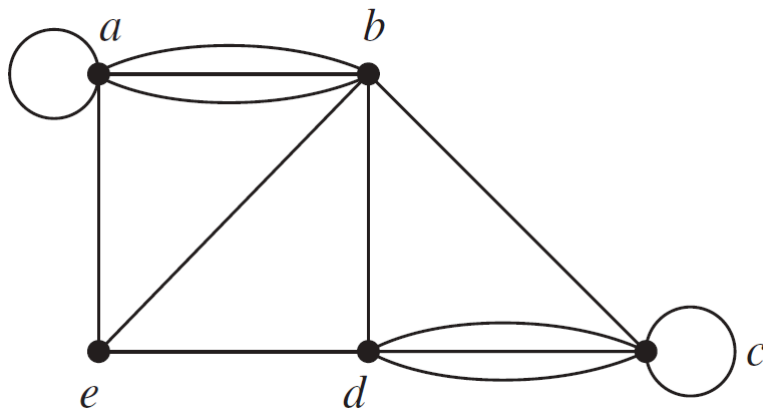
Construct the call graph for a set of seven telephone numbers 555-0011, 555-1221, 555-1333, 555-8888, 555-2222, 555-0091, and 555-1200 if there were three calls from 555-0011 to 555-8888 and two calls from 555-8888 to 555-0011, two calls from 555-2222 to 555-0091, two calls from 555-1221 to each of the other numbers, and one call from 555-1333 to each of 555-0011, 555-1221, and 555-1200.

Question 6.

- a) Explain how graphs can be used to model electronic mail messages in a network. Should the edges be directed or undirected? Should multiple edges be allowed? Should loops be allowed?
- b) Describe a graph that models the electronic mail sent in a network in a particular week.

Question 7.

Find the number of vertices, the number of edges, and the degree of each vertex in the following undirected graph. Identify all isolated and pendant vertices.

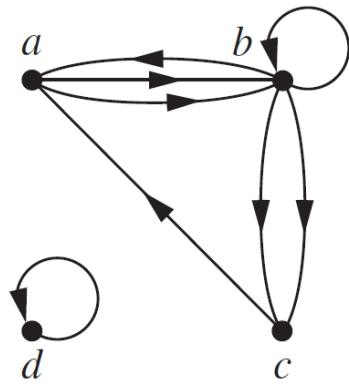


Question 8.

Is it possible to have 3 vertices with vertex degrees equal to 17 and the remaining vertex degrees are even in an undirected graph? Provide formal arguments to justify your answer.

Question 9.

Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the following directed multigraph.



Question 10.

Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.

Question 11.

Recall that Hall's theorem provides a necessary and sufficient condition for complete matching in a bipartite graph $G=(V_1 \cup V_2, E)$. Answer the following questions about this theorem with detailed reasoning.

- A. According to this theorem, if $|A| \leq |N(A)|$ for every $A \subseteq V_1$, then there is a complete matching. In the proof it only considered two cases; case 1: every set of j vertices has at least $j+1$ neighbors and case 2: there exists a set of j vertices that have j neighbors. Why doesn't the proof need to consider the case that every j vertices have less than j neighbors?
- B. In case 1, what problem will you face if instead of removing 1 element out of j elements, you remove 2 elements?
- C. In case 2, we match a set of j vertices with j neighbors and remove the matched vertices, then we say we should be able to apply IH on the rest of the graph. Otherwise, we show a contradiction. Why can't we apply the same technique to case 1?

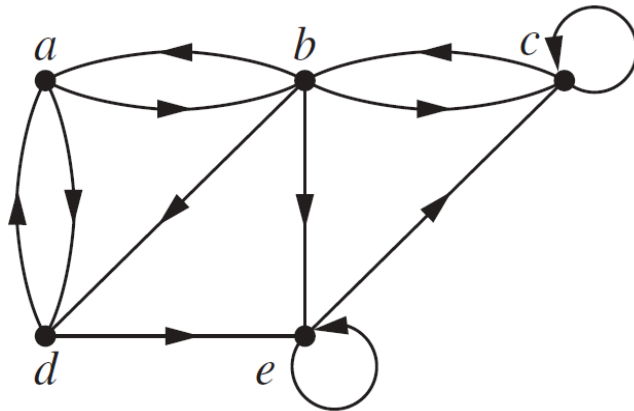
Question 12.

Suppose that a new company has five employees: Zamora, Agraharam, Smith, Chou, and Macintyre. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zamora could do planning, sales, marketing, or industry relations; Agraharam could do planning or development; Smith could do publicity, sales, or industry relations; Chou could do planning, sales, or industry relations; and Macintyre could do planning, publicity, sales, or industry relations.

- A. Model the capabilities of these employees using a bipartite graph.
- B. Find an assignment of responsibilities such that each employee is assigned one responsibility.
- C. Is the matching of responsibilities you found in part B a complete matching from employees to responsibilities? Is it a maximum matching?

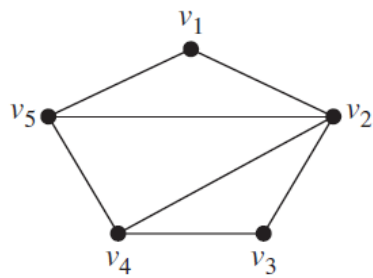
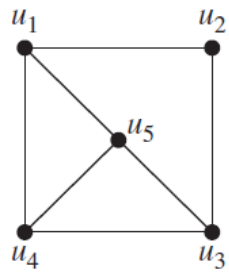
Question 13.

Use an adjacency list to represent the following graph.



Question 14.

Determine whether the following pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.



Question 15.

Determine whether the following pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.

